

Verification and Validation Report: Software Engineering

Team 11, technically functional

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1 Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

2 Symbols, Abbreviations and Acronyms

symbol	description
T	Test

[symbols, abbreviations or acronyms – you can reference the SRS tables if needed —SS]

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3 System testing

The tests outlined below have been divided into 3 key areas:

1. **User registration suite:** the tests listed here have been designed for the purpose of testing the system's handling of the registration process.
2. **PT specific testing:** These tests have been tailored to test the functionalities available to the physiotherapists.
3. **Patient focused testing:** The results from this will give insight into how well the system accomplishes its intended purpose.

Note: Automatic tests will be run using a script. See Appendix for an example template of script. The use of 'display' indicates both audio and visual formats are provided.

3.1 User registration suite

This section contains: user_acc_create, invalid_format_create, cannot_verify_creds, acc_exists.

Table 1: Software Test Cases for User Account Creation

Test Case ID	user_acc_create
Control	Automatic
Rationale	The system must be able to create and store a new account.
State	User does not have an account on the application.
Input	Govt id (name, date of birth), license number (PT) OR invite code (patient), email, password (will be set at the end of the registration process)
Output	"Account created successfully" displayed on page and user is successfully added to the user database
How the test will be performed	The test will be conducted with the use of a script (see template in Appendix)

Test Case ID	invalid_format_create
Control	Automatic
Rationale	The data input by the user has to be in the correct format for our system to accept it as valid.
State	User does not have an account on the application.
Input	Incorrect format of license number/invite code OR incorrect format of date of birth, email, password (will be set at the end of the registration process)
Output	System displays 'incorrect format of one or more inputs'
How the test will be performed	The test will be conducted with the use of a script (see template in Appendix)
Test Case ID	cannot_verify_creds
Control	Automatic
Rationale	Successful verification for both types of users is needed to use the application
State	User does not have an account on the application.
Input	Incorrect license number/mismatch of license number and govt id OR incorrect format of invite code, email, password (will be set at the end of the registration process)
Output	System displays 'License number cannot verified' OR 'invalid invite code'
How the test will be performed	The test will be conducted with the use of a script (see template in Appendix)
Test Case ID	acc_exists
Control	Automatic
Rationale	New users should not already have an account on the system
State	Account already exists on system
Input	Govt id (name, date of birth), license number/invite code, email, password
Output	'Account already exists. Wrong password. Try again.'

How the test will be performed	The test will be conducted with the use of a script (see template in Appendix)
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3.2 PT focused testing

This section contains: PT_sends_invite, PT_adjusts_exercise, PT_deletes_patient_acc.

Table 2: Software Test Cases for Physical Therapist (PT) Functions

Test Case ID	PT_sends_invite
Control	Automatic
Rationale	A patient can only sign up if they have an invite code from the PT.
State	Patient does not have an account on the application while PT does.
Input	PT adds patient's information and contact method (where the code will be sent).
Output	An 'Invite sent' message is displayed on the screen
How the test will be performed	The test will be conducted with the use of a script (see template in Appendix)
Test Case ID	PT_adjusts_exercise
Control	Automatic
Rationale	The system allows the PT to obtain an overview of patient performance and adjust different parameters accordingly (eg. number of repetitions).
State	PT and patient have an account on application, PT has patient on their patient list
Input	PT accesses the patient's profile page and adds/modifies exercise
Output	Patient exercise criteria is modified.

How the test will be performed	The test will be conducted with the use of a script (see template in Appendix)
Test Case ID	PT_deletes_patient_acc
Control	Automatic
Rationale	This ensures PTs control over patients' accounts. PT can remove patient(s) profiles from the application.
State	PT and patient have account on application, PT has the patient on their patient list
Input	PT requests a specific patient's account deletion
Output	Account deleted from user database and PT patient list. Patient is unable to log in ('Account not found').
How the test will be performed	The test will be conducted with the use of a script (see template in Appendix)

3.3 Patient focused testing

This section contains: system_exercise_overview, system_record_exercise, system_performs_correctly, video_store, video_retrieval, post_ex_patient_feedback, incorrect_form_adjustment

Table 3: Software Test Cases for Exercise System and Additional Functions

Test Case ID	system_exercise_overview
Control	Automatic
Rationale	In order to record the patient performing the exercise, the system has to be able to provide instructions to the patient on how to perform it.
State	Patient is logged into the application and has it open on their device. Patient has selected the exercise they want to perform on their exercise list.

Input	Patient clicks on exercise from exercise list.
Output	System displays an overview of the steps required for the exercise.
How the test will be performed	The test will be conducted with the use of a script (see template in Appendix)
Test Case ID	system_record_exercise
Control	Manual
Rationale	Patient performance of the exercise has to be successfully recorded to aid system analysis of its correctness.
State	Patient is currently logged in and has the application open on their device. Patient has selected the exercise they would like to perform from the exercise list.
Input	Patient clicks on the 'record' button
Output	Exercise successfully recorded, saved and ready for viewing
How the test will be performed	• Patient clicks on the 'exercise list' and selects the exercise they would like to perform • System directs patient to a screen with a live video stream of themselves • Patient clicks on the 'record' button • System records exercise
Test Case ID	system_performs_correctly
Control	Manual
Rationale	System has to effectively perform analysis on patient exercise form to provide feedback.
State	Patient is recording exercise
Input	System receives visual of patient performing exercise
Output	System provides feedback to patient (e.g. instructions for the next steps in the exercise)

How the test will be performed	• Patient performs exercise correctly • System monitors and analyzes exercise performance in real-time • System provides correct feedback
Test Case ID	video_store
Control	Automatic
Rationale	The system has to be able to store a recorded video of the patient performing the exercises.
State	Patient has performed and recorded exercise being performed.
Input	Patient clicks on 'Save exercise' button. <i>Note: The team has not decided whether to store the video locally or on the user database.</i>
Output	Video has been successfully saved and is ready for viewing.
How the test will be performed	The test will be conducted with the use of a script (see template in Appendix)
Test Case ID	video_retrieval
Control	Automatic
Rationale	The system has to be able to retrieve the most recent recorded video in storage (either locally or database).
State	Patient has performed and recorded exercise being performed
Input	Patient OR PT clicks on video stored (local or database) for viewing. <i>Note: The team has not decided whether to store the video locally or on the user database.</i>
Output	Video has been successfully retrieved and is ready for viewing.
How the test will be performed	The test will be conducted with the use of a script (see template in Appendix)
Test Case ID	post_ex_patient_feedback
Control	Automatic

Rationale	The system has to be able to store patient feedback of the performed exercise (e.g. the level of (perceived) difficulty of the exercise)
State	Patient has performed and recorded exercise being performed and system has successfully performed analysis of the performance
Input	Patient input into feedback form (See Appendix <i>Patient post-exercise feedback form</i>).
Output	Answers are stored successfully
How the test will be performed	The test will be conducted with the use of a script (see template in Appendix)
Test Case ID	incorrect_form_adjustment
Control	Manual
Rationale	The system has to be able to differentiate between correct and incorrect forms to provide accurate feedback.
State	Patient is performing the selected exercise
Input	Incorrect form during exercise performance
Output	Adjustment directives displayed to patient
How the test will be performed	• Patient performs exercise with incorrect form • System detects the incorrect posture/movement • System provides real-time adjustment instructions

4 Nonfunctional Requirements Evaluation

Table 4: Non-Functional Requirements Testing

Test Case ID	ui_testing
Type	Static
Rationale	Ensure user interface meets usability standards and stakeholder expectations
State	N/A
Input	User interface elements and survey responses
Output	Modified user interface based on feedback
How the test will be performed	• A survey will be given to users (see 'Stakeholders' in SRS for the type of users) to test different interface elements • See Appendix for survey questions • Based on the feedback received, the user interface will be modified in order to satisfy additional requirements
Test Case ID	ex_int_speed
Type	Automatic
Rationale	Ensure exercise interface loads within acceptable time limits for good user experience
State	User is logged in and has application open on their device
Input	User clicks on exercise interface
Output	Interface loads within 2 seconds
How the test will be performed	The test will use external tools (Postman) to measure API latency. In order to replicate real world conditions, an increasing amount of simulated user loads can be used. Statistical analysis will be performed on the collected results to provide a more accurate value.
Test Case ID	fail_recovery
Type	Dynamic

Rationale	The system must be able to restore its pre-failure state 95% of the time within 5 minutes
State	User is currently using the application on their device
Input	System failure (simulated)
Output	System restarts and recovers within 5 minutes
How the test will be performed	• A failure scenario will be simulated • Recovery time and success rate will be measured • System's state will be compared to pre-failure state to confirm if properly restored.
Test Case ID	Interrupted_vid
Type	Dynamic
Rationale	The system must be able to attempt resubmission of video after restoring internet connection
State	A video of the user performing the exercise is being uploaded
Input	Internet connection interruption during upload
Output	Video is successfully re-queued for upload to the system. Re-upload is resumed at the point of failure
How the test will be performed	• Network is disconnected during video upload • Connection is restored after interruption • The system detects interruption and queues video for re-upload • Confirm upload resumes from point of failure (not restarting) • Video integrity is confirmed after successful upload
Rationale	PR-RFT 3, PR-RFT 4, PR-CR1, PR-SER 1 These requirements address advanced system capabilities beyond current testing scope

How the test will be performed	• These requirements are unable to be tested in a timely manner due to project constraints (time, money) • Testing deferred to future development phases • Requirements marked for re-evaluation in next project iteration
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5 Comparison to Existing Implementation

This section will not be appropriate for every project.

6 Unit Testing

7 Changes Due to Testing

[This section should highlight how feedback from the users and from the supervisor (when one exists) shaped the final product. In particular the feedback from the Rev 0 demo to the supervisor (or to potential users) should be highlighted. —SS]

8 Automated Testing

9 Trace to Requirements

10 Trace to Modules

11 Code Coverage Metrics

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Reflection.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which parts of this document stemmed from speaking to your client(s) or a proxy (e.g. your peers)? Which ones were not, and why?
4. In what ways was the Verification and Validation (VnV) Plan different from the activities that were actually conducted for VnV? If there were differences, what changes required the modification in the plan? Why did these changes occur? Would you be able to anticipate these changes in future projects? If there weren't any differences, how was your team able to clearly predict a feasible amount of effort and the right tasks needed to build the evidence that demonstrates the required quality? (It is expected that most teams will have had to deviate from their original VnV Plan.)